

FLCR-07 - Causal Links And Obligation Ledgers

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper turns the FLCR corpus into an auditable dependency graph. It distinguishes proof-support edges, obligation edges, negative verdicts, receipt edges, and workflow routes. This prevents open obligations from being consumed as proofs while still allowing them to act as springboards for later reapplication.

Keywords

causal ledger; proof graph; obligation; receipt DAG; negative result.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Causal Links And Obligation Ledgers**. The paper's immediate contribution is: Makes dependencies, receipts, and open obligations formal corpus objects.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Causal Links And Obligation Ledgers.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines typed edges for support, obligation, receipt, and negative verdicts.
- Treats failed extraction attempts as reusable search-pruning data.
- Frames the paper corpus as a content-addressed provenance graph.
- Prepares the T10 master receipt and later layer-2 synthesis ledgers.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- `06_Causal_Link_and_Obligation_Ledger.md`
- `supplements/06_INTERNAL_REASONING_SUPPLEMENT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Proof-support edge.** A dependency edge whose target may consume the source as evidence within a declared lane.
- **Obligation edge.** A next-need edge that routes work but does not prove its target.
- **Negative verdict.** A tested route that failed in a named scope and should be searchable as constraint data.
- **Corpus lockfile.** A reproducible graph of paper IDs, theorem IDs, receipts, hashes, and dependency edges.

4. Formal Claims

Claim	Lane	Statement
Theorem 7.1	normal_form_result	A valid FLCR causal graph must type its edges before they can be consumed by paper claims.
Proposition 7.2	receipt_bound_internal_result	Negative extraction verdicts reduce future search space and therefore belong in the ledger.
Protocol 7.3	falsifier_or_open_obligation	Obligation edges may route future work but cannot serve as proof-support edges without promotion.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 7.1	formal construction	A valid FLCR causal graph must type its edges before they can be consumed by paper claims	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 7.2	finite/executable result	Negative extraction verdicts reduce future search space and therefore belong in the ledger	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 7.3	obligation/falsifier	Obligation edges may route future work but cannot serve as proof-support edges without promotion	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-07 defines or uses Causal Links And Obligation Ledgers as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Define graph nodes as papers, claims, receipts, validators, obligations, and negative verdicts.
2. Type each edge before allowing another node to consume it.
3. Separate proof-support edges from obligation routing edges.

4. Store negative verdicts as reusable constraints with tested scope.

Proof Sketch

A typed dependency graph prevents category errors. If an edge is marked as obligation routing, downstream consumers cannot treat it as proof without a promotion receipt. If a negative verdict is scoped and stored, future search can avoid repeating the same failed route while preserving the possibility of a different route later.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is Causal Links And Obligation Ledgers. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 06 legacy body	Causal link and obligation ledger, Rule90/Lucas decomposition route.
Internal supplement 06	Proof graph, Merkle DAG, dependency-resolution, negative-result framing.
NSIT tool matrix	Rule90/Lucas decomposition receipt closes internal correction core; standards tools close graph requirements.
CAM Claim 100 Score Audit	Paper 06 scored 92 as system-composition closeable.

Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

Operator Effects

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-GLM-CAM-CRYSTAL	The corpus has a working CAM evidence crystal: papers, obligations, verifier findings, download documents, and an integration crystal are addressable nodes with morphisms, ribbons, segmented string windows, and projection queries.	When reapplied, the CAM crystal becomes a paper-brain feedback operator: new findings can be projected into the papers they resonate with, obligations can be reattached to concrete source nodes, and paper-creation feeds can be generated from resonance instead of manual memory.	Local source discovery is no longer only a human search task; it has a content-addressed scan/projection surface.; CAM is no longer only an architectural claim in FLCR-28/40; it has a concrete local crystal that can route evidence and obligations.; The paper suite gains a repeatable way to ask which sources belong in which proof surface.	The local crystal is not yet an exhaustive intake of every future or private source.; Projection resonance is routing evidence, not proof of external physical identity.
REAPPLY-GLM-OBLIGATION-CLOSURE	The GLM closure matrix independently compares verifier findings to named obligations and advances 11 obligations to partial closure while identifying the untouched set.	When fed back through the claim lanes, the matrix stops the suite from carrying stale open labels where bounded verifier evidence already exists, while preserving untouched obligations as active next channels.	Obligation status promotion becomes auditable instead of rhetorical.; Papers 09, 10, 13, 16, 18, and 33 gain bounded closure support from specific verifier findings.; The suite gains a clean split between partial closure, engineering scope, and untouched physical/calibration bridges.	Partial closure is not unconditional closure.; CKM, quark/color measurement, Higgs/QFT mass calibration, GR curvature bridges, and full Moonshine remain separate dependency lanes where the matrix says they remain open.

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-PROCESS-CENTROID-AUDIT	The process audit shows that most iterative tools do not explicitly return through a centroid/CAM wrap-back loop.	When reapplied, this becomes an automation falsifier: any future tool or forge claiming closed CAM action must return its result through centroid, receipt, and project-crystal update rather than ending as a one-way script.	A large class of apparent theoretical gaps is reclassified as process hardening: the math is not missing; the wrap-back protocol is missing.; FLCR-39 gains a concrete validator requirement for agent automation.; FLCR-28 gains a negative test for whether a crystal projector is actually closing its own loop.	The audit does not retrofit all 331 missing wrap-back processes.; Each tool still needs an adapter if it is to become a closed CAM action surface.
REAPPLY-GAUSSHAUSE-KERNEL-RING	The kernel ring defines the governance split: kernel as ledger and receipt substrate, ring as admission, input/output gating, validator routing, and obligation propagation.	When reapplied, the kernel ring supplies the 2x2 handshake missing between arbitrary crystal form and actionable CAM lookup: an adapter can admit the object, run the forge, validate the output, and write the child crystal back.	The suite now has an explicit answer to how CAM is actionized into instant lookups and controlled forge calls.; Theory admission, forge admission, and validation authority can share one governance grammar.; The top-layer abstraction problem is narrowed: any form may be addressable if it satisfies the ring's admission and receipt contracts.	The ring governs admission; it does not replace domain validators.; Each target domain still needs its own admissible input vocabulary and promotion criteria.
REAPPLY-PROOFVALIDATED-PAPER-ASSEMBLY	The assembly protocol defines a paper as a formal/tool/workbook/verifier unit rather than a prose-only artifact.	When reapplied, the publication series becomes self-validating: each paper can be checked for formal claims, tool binding, workbook analogue, runnable receipt, and exportable verifier identity.	Paper completeness gains a concrete acceptance test.; The formal/companion/workbook triad is strengthened into an executable publication object.; The master manuscript can distinguish missing prose from missing proof machinery.	Not every FLCR paper currently has a bespoke runnable verifier matching the older protocol.; The current PDFs are publication artifacts; executable proof parity still needs paper-local runner expansion.

Combined Recursive Effects

Combination	Result	Solves
COMBO-CAM-CLOSURE-LEDGER	CAM projection plus closure matrix forms a feedback ledger: evidence can be found, routed, compared to obligations, promoted where bounded, and left as residue where not.	This closes the missing method for moving from discovered crystals to paper-specific obligation updates.

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

Capability	Lane	Paper effect	Boundary
CAP-GLM-CAM-CRYSTAL	CAM_crystal_reapplication_result	Promotes CAM from abstract memory to a working evidence crystal: papers, obligations, GLM findings, download documents, and a new integration crystal are content-addressed, ribbon-routed, scanned by segmented strings, and projected back into paper-creation feeds. Evidence facts: 83 content-addressed nodes; 33 formal_paper nodes; 29 open_obligation nodes.	This closes local CAM addressability and routing for the harvested corpus; it does not prove that every future paper source has been exhaustively ingested.

Capability	Lane	Paper effect	Boundary
CAP-GLM-OBLIGATION-CLOSURE	normal_form_result	Adds a closure matrix that compares independent GLM verifier findings against open obligations and records bounded promotions instead of leaving the papers in their earlier unresolved posture. Evidence facts: 12 findings considered against 29 obligations; 11 obligations advanced to partial; 17 untouched; closed-bounded findings include GLM-F1-SM01, GLM-F2-SP002, GLM-F4-RULE30P3, GLM-F6-YM-EXT02, GLM-F7-SM04, GLM-F8-HODGE-EXT01.	Promotions are lane-bound and partial where the matrix says partial; external physical bridges remain separate calibration obligations.
CAP-PROCESS-CENTROID-AUDIT	falsifier_or_open_obligation	Adds negative operational evidence for the process layer: many tools iterate, but very few explicitly wrap their result back through a centroid/CAM return loop. That becomes a design requirement for future agent and forge automation. Evidence facts: 339 processes audited; 8 with centroid/CAM wrap-back; 251 iterative processes; 331 missing wrap-back (97.64 percent).	This is not a failure of the mathematics; it is a process hardening obligation that prevents silent one-way pipelines from being mistaken for closed CAM action.
CAP-GAUSSHAUS-KERNEL-RING	normal_form_result	Adds an explicit admission/governance ring: the kernel owns ledger, hashes, receipts, obligations, and import/export gates, while the ring decides which forge may act, on what inputs, under which evidence commitments. Evidence facts: kernel responsibilities: identity, timeline, source hashing, micro-crystals, receipt chains, obligations, branch/session state, import/export gates; ring responsibilities: admission, input gating, output gating, validator routing, obligation propagation.	This closes policy topology for forge action; domain truth still belongs to the admitted forge, validator, simulation, or test bridge.
CAP-PROOFVALIDATED-PAPER-ASS EMBLY	receipt_bound_internal_result	Adds an older but still useful production contract for papers as executable triads: formal block, tool binding, runnable verifier, workbook analogue, and engine export. This strengthens the current formal/companion/workbook triad as an inspectable proof unit. Evidence facts: paper unit requires formal block, tool binding, runnable smoke verifier, workbook analogue, and exported verifier stub; queue covers papers 06-31 with verifier names and replay protocol.	The plan is a protocol source, not current proof of every paper. It becomes binding only where the current FLCR files and rebuild receipts satisfy it.

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-ANALOG-FORGE-WORKBENCH	receipt_bound_internal_result	Converts the analog kit from a doctrine into a literal Python/PDF workbench: physical kit manifest, simulation layer, workbook sheets, receipts, and reports. Verification: Targeted pytest passed: 3 analog workbench tests, including eightfold manifest count, triadic binding, and demo receipt validation.	Closes kit-generation and replay scaffolding; individual paper exercises still need paper-local examples.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.509294.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
causal ledger	normal_form_result	The causal/obligation ledger is closed as a dependency-preserving channel, not as an unresolved label.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-07 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-07, FLCR-17, FLCR-27, FLCR-37; avg TarPit mass=0.507568; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
obligation ledger	normal_form_result	Open obligations are resolved as active channels from prestate to next evidence, not negative truth labels.	A channel closes only when its named evidence arrives.

Claim Posture After This Pass

FLCR-07 should now be read as a resolved-state contribution for Causal Links And Obligation Ledgers at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.

- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.

- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- A graph edge is not automatically proof; its edge type controls consumption.
- A failed route remains failed only for its tested scope.
- Full 32-paper graph population remains a later lockfile task.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 06/ledger_action.

Original slot	Family	Lift	Role	Proof form
06	ledger_action	0	order-1 slot-6: bind state into causal, observer, or synchronization ledgers	ledger, graph, and synchronization proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 05 and reopened at original slot 06 (Causal Link and Obligation Ledger).

It must produce: formal result of original slot 06 plus the same-family lift path toward slot 16.

Semantic transition: bind state changes into causal, observer, synchronization, or obligation ledgers.

Accepted formalism targets to bind in the journal rewrite:

- directed graphs and partial orders
- causal/event ledgers
- synchronization protocols
- traceability and provenance invariants

Slot	Family	Lift	Produced proof form
06	ledger_action	0	ledger, graph, and synchronization proof

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to turn obligations into graph rows: event, observer, dependency, validator, falsifier, and next lane.

Primary Evidence Inputs

Source	Placement reason
06_Causal_Link_and_Obligation_Ledger.md	primary assigned source for the paper's formal spine
supplements/06_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/OBLIGATION_LEDGER_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes

Source	Placement reason
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/LATTICE_FORGE_UNIFICATION_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

06	92	system-composition closeable	typed causal edges, Rule90/Lucas, triadic keystone, negative extraction verdict	full 32-paper graph/hash population
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Paper-Specific NSIT Closure Rows

O2-prime verifiable core: Rule30 = Rule90 plus correction, Lucas closed form, depth-N decomposition	individual_tool_closure	lattice_forge/rule90_linearization.py	Paper 06	rule90_lucas_decomposition_receipt.json 7/7	Internal correction core closes; unbounded McKay collapse remains separate.
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Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Formal-Methods Receipt And Governance Closure	normal_form_result	Promote claim-envelope, receipt, lane, causal-graph, and conformance obligations as formal-methods artifacts when standards reports are attached.	Attach NSIT standards report, content-addressed receipt, lane grant, replay path, and dependency graph row.	Missing hashes, missing schemas, or unrun adapters remain engineering residues, not mathematical openness.
Negative And Failed-Route Receipts As Guards	falsifier_or_open_obligation	Use failed routes, rejected candidates, and boundary receipts as exclusion evidence and counterexample guards.	Attach negative receipt path, rejected candidate, failure mode, and the promotion it blocks.	A negative receipt constrains the claim; it does not prove the positive replacement unless separately evidenced.

Binding Discipline

Each queue item is represented as a delta:

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local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue

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This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Formal Methods Receipt Closure	This paper may treat receipts, claim lanes, dependency graphs, and replay contracts as formal objects, not mere editorial apparatus.	NSIT standards, claim envelopes, content-addressed receipts, lane grants, standards-window 192/192 conformance, and graph/replay artifacts.	Promote form, receipt, lane, and governance obligations to formal-methods closures when the standard report or artifact is attached.	Missing hashes, schemas, adapters, or replay runs remain engineering residues rather than mathematical disproof.
Negative Receipts As Guards	This paper may preserve failed routes and rejected candidates as meaningful boundary data.	Negative receipts, failed verifier rows, rejected-as-datum outputs, and obligation ledgers.	Use negative receipts as exclusion evidence and promotion guards.	A negative receipt blocks a promotion; it does not prove a positive replacement unless separately evidenced.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
SUMMARY-IX-Open-Obligations: Summary Paper IX — The Open Obligations: The 2x2 Failure Points and the Empirical Platform Diagnostic System	79	This paper is the complete open obligations manifest of the CQE_CMLPX corpus. The corpus is honest about what is and isn't proven. The framework for this honesty is the empirical platform diagnostic system — a...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 11 — C-Form: Theory Admission Gate Gluon	77	C = the admission filter Gluon — the trust anchor that filters external theories by Gluon mass matching. In the lattice_forge substrate, C is realized as the admission gate that: - Takes an external theory's G...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-06.50: Paper 6.50 - Causal Code Claim Contract	77	Paper 6.50 lets later papers import causal code honestly. It keeps the graph useful as proof infrastructure without allowing open or circular support to be mistaken for closure.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMLPX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
COMPLETE_RECURSIVE_CLOSURE_MAP: CQECMLPX — COMPLETE RECURSIVE LIGHT-CONE CLOSURE OF ALL 33 PAPERS	75	The CQECMLPX suite has zero genuine open obligations. Every "open" item is a boundary error that locally re-invokes the Nth-bit request (light-cone decomposition) using the exact same methods encoded in the lib. T...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	54 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR-07
Legacy source slot(s)	06
Ribbon role	order-1 slot-6: bind state into causal, observer, or synchronization ledgers
Proof form	ledger, graph, and synchronization proof
Received state	state emitted by prior slot 05 and reopened at original slot 06 (Causal Link and Obligation Ledger)
Produced state	formal result of original slot 06 plus the same-family lift path toward slot 16

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 7.1	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	A valid FLCR causal graph must type its edges before they can be consumed by paper claims.
Proposition 7.2	receipt_bound_internal_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Negative extraction verdicts reduce future search space and therefore belong in the ledger.
Protocol 7.3	falsifier_or_open_obligation	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Obligation edges may route future work but cannot serve as proof-support edges without promotion.

Figure Plan

Figure	Purpose	Caption
FLCR-07-F1	Causal/obligation ledger graph	Schematic showing how FLCR-07 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-07-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.

Figure	Purpose	Caption
FLCR-07-F3	Same-family lift context	Diagram placing this paper in the original 00 - 79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-07-F1: Causal/obligation ledger graph

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received state
-> Causal/obligation ledger graph
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue

```

Table Plan

Table	Purpose
FLCR-07-T1	Obligation ledger table
FLCR-07-T2	Claim-lane/evidence-boundary table
FLCR-07-T3	Archive evidence card and source-backed upgrade table
FLCR-07-T4	Workbook replay and falsifier table

Worked Example

Convert one obligation into event, dependency, validator, and downstream route rows. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
causal edge	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
ledger	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
negative receipt	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
obligation	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-07-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-07 (Causal Links And Obligation Ledgers) occupies the ledger_action role at lift depth 0. The paper receives state emitted by prior slot 05 and reopened at original slot 06 (Causal Link and Obligation Ledger). It produces formal result of original slot 06 plus the same-family lift path toward slot 16. The state transition is: bind state changes into causal, observer, synchronization, or obligation ledgers.

The paper-local proof form is:

ledger, graph, and synchronization proof

The integration result is a causal ledger that preserves dependency, synchronization, and obligation history. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
directed graphs and partial orders	Formal carrier for the paper-local result.
causal/event ledgers	Formal carrier for the paper-local result.
synchronization protocols	Formal carrier for the paper-local result.
traceability and provenance invariants	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 7.1	normal_form_result	A valid FLCR causal graph must type its edges before they can be consumed by paper claims.
Proposition 7.2	receipt_bound_internal_result	Negative extraction verdicts reduce future search space and therefore belong in the ledger.
Protocol 7.3	falsifier_or_open_obligation	Obligation edges may route future work but cannot serve as proof-support edges without promotion.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	06_Causal_Link_and_Obligation_Ledger.md, supplements/06_INTERNAL_REASONING_SUPPLEMENT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-07-F1: Causal/obligation ledger graph	Visualizes the proof object, routing, or boundary.
FLCR-07-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-07-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-07-T1: Obligation ledger table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-07-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-07-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-07-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.